

Market Making Alpha: How Virtu Won 1,237 Days — and Why Jane Street Keeps Ending Up in Court

Court filings, SEC administrative orders, peer-reviewed papers, and on-record executive interviews reveal something most finance professionals never piece together: the bid-ask spread is not a fee. It is the residual of an information war, and the firms that win it do so through three separable, documented mechanisms — fair-value forecasting accuracy, real-time adverse selection filtering, and inventory velocity. The firms that have built these three levers into systematic processes at scale have produced the most reliably profitable trading businesses in modern financial history.



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Why the Spread Exists: The Glosten-Milgrom Proof

The foundational misconception in most market making analysis is that the bid-ask spread compensates the market maker for bearing inventory risk or processing costs. The 1985 proof by Lawrence Glosten and Paul Milgrom demolished that assumption four decades ago, and almost every subsequent advance in market making practice rests on what they showed.

Bid, Ask, and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders, published in the *Journal of Financial Economics*, demonstrated that a positive bid-ask spread emerges even when the market maker is risk-neutral and earns zero expected profit. The abstract states the core result plainly: the spread exists because of traders with superior information. A risk-neutral specialist who breaks even in aggregate must still charge a spread — not to profit, but to recover from uninformed traders what is systematically lost to informed ones.

The model divides counterparties into two populations: informed traders who know the asset's true value and trade profitably against the market maker; and uninformed liquidity traders who trade for exogenous reasons — hedging, rebalancing, cash needs. The market maker cannot distinguish them before a trade. The rational equilibrium spread is the one at which profits from uninformed flow exactly offset expected losses to informed flow. The spread is not a margin — it is a break-even condition under asymmetric information.

The practical implication is precise and actionable: every dollar of market making alpha attributable to information asymmetry must come from one of two sources — either (a) identifying informed flow before it trades, or (b) forecasting fair value with sufficient accuracy that even informed flow does not know more than the model. A third operational dimension — inventory velocity, governing how quickly a taken position is neutralized regardless of how it was priced — operates alongside these two and is separable from them.

These three dimensions are the entire framework. Everything that follows in this article is a real-world implementation of one or more of them, drawn from primary sources that are publicly available and legally binding.

The Virtu Record: What the SEC Filing Actually Says

The most-cited statistic in market making — Virtu Financial's near-perfect daily trading record — is almost always reported with imprecision. The [SEC-filed S-1 prospectus](#) states it exactly: "we had only one losing trading day during the period depicted, a total of 1,238 trading days" from January 1, 2009 through December 31, 2013. That means 1,237 profitable days and one losing day across five years.

But the filing's deeper content is more instructive than the headline figure. The same document discloses that in 2013, Virtu earned 27% of its

Adjusted Net Trading Income from Americas equities, 11% from EMEA equities, 11% from APAC equities, 23% from global commodities, 20% from global currencies, and 9% from options, fixed income, and other securities. No single category exceeded 30% of total income. This diversification is not incidental — it is the structural source of the near-zero loss-day probability. With thousands of uncorrelated microstructure edges running simultaneously, the law of large numbers ensures that no single position-level adverse move can aggregate into a firm-level negative day.

A separate [SEC correspondence filing \(CORRESP\)](#) submitted during the IPO process makes Virtu's actual trading mechanism explicit. Responding to SEC staff questions, Virtu described three discrete strategies. The first is "single instrument market making": actively quoting a single instrument "with the intention of profiting by capturing the spread between the bid and offer," placing bids and offers at or near the inside market simultaneously. The second is "multi-leg market making": simultaneously quoting related instruments — for instance, an ETF and its underlying basket — and profiting from the relative spread between economically linked products. The third is "hedging" — using related instruments to neutralize price risk on accumulated inventory.

The [2015 S-1 re-filing](#) added a critical disclosure: "for the 252 trading days of 2014, we averaged approximately 5.3 million trades per day globally across all asset classes, and we profitably exited 49% of our overall positions." Fewer than half of Virtu's individual trades were

profitable. The CORRESP filing confirmed the 2014 Q1 figure: 49.6% of positions were unprofitable. The alpha is not in picking winners trade-by-trade. It is in the mathematical expectation of a 50.4% win rate multiplied 5.3 million times daily, diversified across [more than 11,000 securities and 225 unique exchanges in 35 countries](#).



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The Inventory Control Framework: Avellaneda-Stoikov in Practice

The Virtu record establishes that diversification and win-rate mathematics produce the aggregate edge. But that aggregate conceals a granular operational problem: every individual filled order creates a position, and an accumulating inventory is directional exposure by another name. For any market maker holding a position longer than a millisecond, inventory

management is not a secondary concern — it is the central technical problem. The canonical academic solution was provided by Marco Avellaneda and Sasha Stoikov in [*High-Frequency Trading in a Limit Order Book*](#), published in *Quantitative Finance* (Volume 8, Issue 3, 2008).

The paper formalizes an insight that every experienced market maker knows intuitively: when a market maker's inventory is long, they should skew their ask tighter and their bid wider — making it cheaper to sell to them and more expensive to buy from them — to attract sellers and reduce the inventory. The model achieves this through a "reservation price," the market maker's private valuation of the asset given current inventory:

$$r = S - q \cdot \gamma \cdot \sigma^2 \cdot (T - t)$$

where S is the mid-price, q is the inventory position (positive = long, negative = short), γ is the risk aversion parameter, σ^2 is the asset's variance, and $(T - t)$ is time remaining in the trading session. When $q > 0$ (long inventory), the reservation price drops below mid — the market maker's private valuation reflects their urgency to offload. Bid and ask quotes are then set symmetrically around this private valuation rather than around the market mid-price, skewing quotes to attract the offsetting flow. As the session approaches close, $(T - t) \rightarrow 0$ and the reservation price converges back to market mid, forcing inventory to flatten.

The practical importance of this framework is that it shows market making alpha is not purely about prediction accuracy. A firm with a mediocre price forecast but superior inventory management — one that never lets accumulated directional exposure compound — can outperform a firm with excellent forecasts but poor inventory discipline. The two dimensions are separable.

This is Lever 3 — inventory velocity — operating in parallel with the two information-asymmetry levers that Glosten-Milgrom identified. The five case studies that follow are each, at root, implementations of one or more of these three mechanisms. The [2012 Fodra-Labadie extension on arXiv](#) adds directional bets and inventory penalties at horizon, extending the framework to the conditions real desks actually face.

XTX Markets: A Decade of On-Record Evidence

XTX Markets provides the most extensively documented case of what prediction-quality-over-latency market making looks like at scale. The firm was [founded in January 2015](#) by Alexander Gerko, a PhD mathematician from Moscow State University who served as [Head of FX Trading at GSA Capital](#) before spinning XTX out. In 2022, XTX reported [profits of £1.1 billion, a 64% increase on the prior year, on revenues of £2.5 billion](#), making Gerko the [UK's largest individual taxpayer in 2023](#) — paying approximately £664 million in personal tax that year — and

culminating in a record £682 million personal distribution from XTX profits in 2024.

What makes XTX documentable is the directness of its senior leadership's public statements about strategy. In a [February 2017 Risk.net interview](#), co-CEO Zar Amrolia gave the clearest public articulation of the firm's operational philosophy: "We're not a latency-sensitive firm. We take risk and we operate a fair-value model." The same article confirms that XTX holds positions "for more than 10 minutes on average in G10 markets and for 20 minutes in emerging markets" — timeframes that are orders of magnitude longer than pure latency arbitrage, and which require a confident short-term price forecast to be profitable.

The structural implication is important. A firm holding EUR/USD for 10 minutes is not operating as a latency arbitrageur. It is making a conditional prediction about where EUR/USD will trade over the next 10 minutes, updating that prediction continuously as new data arrives, and managing the resulting inventory position while the prediction plays out. If the prediction is wrong more than some threshold percentage of the time, the inventory losses exceed the spread capture and the strategy is unprofitable. XTX's documented profitability is therefore direct evidence that their multi-asset pricing models are, on average, correct at 10-minute horizons — a statement that would be remarkable about any firm.

A second key disclosure came in [December 2017 Risk.net awards coverage](#), when Amrolia described the firm's decision to move to zero hold time on its bilateral FX quotes. Traditionally, FX liquidity providers

used a "last look" window — a holding period of 50 to 200 milliseconds — during which they could cancel a trade after seeing the client's order but before execution. This option protects the market maker from being filled on stale quotes during fast-moving markets. XTX eliminated this window entirely. The result: bilateral volumes with clients rose 50%, and the firm's direct client count grew from 20 to 100 in a single year.

The mechanism the decision reveals: last look is an adverse selection buffer that protects per-trade profitability by allowing the rejection of trades that arrive precisely when the market maker's quotes have become stale. By removing it, XTX accepted worse per-trade economics in exchange for flow volume growth. This trade-off is only rational if the underlying pricing model is accurate enough that last look provides marginal rather than essential protection — a self-referential claim about model quality that the documented bilateral volume growth corroborates.

Amrolia's formal statement on the zero-hold-time decision, quoted in the [Finance Magnates report](#) at the time of announcement: "With recent improvements in technology and market data, we believe that it is hard to justify the use of a latency buffer in pricing. All direct counterparties of XTX Markets now enter into transactions either on a 'Non Last Look' or a 'Zero Hold Time' basis with no latency buffer."

The Adverse Selection Instrument: VPIN at Tudor Investment Corp

The problem every market maker faces — distinguishing informed from uninformed flow in real time — had been formalized academically six years before XTX's zero-hold-time decision, and the primary author was working inside a live hedge fund when he published it.

Marcos López de Prado's institutional affiliation on the [VPIN Flash Crash paper](#) is significant: he is listed as "head of High Frequency Futures at Tudor Investment Corporation in Greenwich, CT." This means the Volume-Synchronized Probability of Informed Trading (VPIN) metric was developed inside an active hedge fund, with access to live trading data, by someone with operational responsibility for high-frequency futures strategies. The paper was published in the [Journal of Portfolio Management \(Vol. 37, No. 2, Winter 2011\)](#).

The mechanism: VPIN divides trading volume into equal-sized "buckets" rather than equal time intervals, classifying each trade as buyer- or seller-initiated using bulk volume classification. The VPIN metric for each bucket is the absolute difference between buy volume and sell volume, divided by total volume. High VPIN — approaching 1 — means nearly all volume in a given period is one-directional, which is the statistical signature of informed, directional trading by agents who believe the asset is moving systematically. Low VPIN means flow is balanced, consistent with uninformed noise trading.

The authors explicitly proposed "a 'VPIN contract' that would allow liquidity providers to dynamically monitor and manage their risks" — acknowledging that the metric's primary application is as a market maker's real-time risk tool, not merely an academic measure. A companion analysis published as an NYU Stern working paper, [*Flow Toxicity and Liquidity in a High Frequency World*](#), argued that VPIN reached its highest level for the full sample on May 6, 2010 — the day of the Flash Crash — and that the elevated toxicity caused market makers to pull liquidity, converting from liquidity providers to liquidity consumers. This claim has been disputed in peer-reviewed literature: Andersen and Bondarenko, writing in the *Journal of Financial Markets* (2014), found that VPIN actually peaked after the Flash Crash rather than before or during it. The operational value of VPIN as a real-time toxicity signal is documented regardless of the dispute over its precise Flash Crash timing.

The operational inference for any firm running a VPIN-type system in live markets: when VPIN exceeds a threshold, widen quotes or pull them entirely; when VPIN normalizes, return to standard spreads. This is the real-time implementation of Glosten-Milgrom's equilibrium condition — adjusting the spread to maintain expected break-even profitability as the proportion of informed traders in the flow changes dynamically.

Citadel Securities: The SEC Administrative Record on Information Asymmetry

The most explicit primary documentation of how a market maker monetizes an information advantage comes not from any firm's voluntary disclosures but from an SEC enforcement action.

On January 13, 2017, the SEC issued [Administrative Proceeding Release No. 33-10280](#) against Citadel Securities LLC. The order found that two algorithms — FastFill and SmartProvide — were activated when Citadel's systems detected discrepancies between the SIP (Securities Information Processor) feed and direct exchange data feeds. The SIP consolidates best bid and offer data from all exchanges; direct feeds deliver the same data faster, before the SIP has published the update. The latency gap between the two is typically measured in microseconds to milliseconds.

The SEC's order is precise about what each algorithm did with this information gap. FastFill immediately internalized a marketable order at the SIP national best bid or offer — the stale price — rather than the better price already visible on the direct feed. SmartProvide did not internalize at the SIP price, nor did it seek to obtain an execution at the best available price by routing to the market. Instead it routed a non-marketable order — one not priced to execute immediately — bypassing the better price the direct feed was already showing.

The SEC [press release](#) quoted then-Acting Enforcement Director Stephanie Avakian: "Citadel Securities made misleading statements suggesting that it would provide or try to get the best prices it saw for retail orders routed by other broker-dealers." Citadel settled for \$22.6 million without admitting or denying the findings.

The enforcement document also quantifies Citadel's market position at the time: the SEC order states that Citadel's processing of retail equity orders "accounts on average for approximately 35% of the average daily volume of retail equity shares traded in the U.S. markets." Roughly one-third of all US retail equity flow was running through algorithms that were using a real-time information advantage over the customers whose orders they were processing. This is the regulatory record of what market making alpha from information asymmetry looks like when it crosses compliance boundaries.

The Jane Street Trade Secret Litigation: A Billion-Dollar Strategy Made Public

Regulatory enforcement is one channel through which market making mechanics enter the public record. Civil litigation is another — and in 2024 it produced the most consequential accidental disclosure in recent market making history.

In April 2024, Jane Street Group filed suit in the United States District Court for the Southern District of New York, [Case 1:24-cv-02783-PAE](#),

against Millennium Management and two former traders, Douglas Schadewald and Daniel Spottiswood. The amended complaint alleged that the two traders, who had been "intimately" involved in developing a proprietary India options strategy at Jane Street, carried that strategy to Millennium upon departing, enabling Millennium to replicate it.

The complaint's financial disclosures were extraordinary. [BNN Bloomberg's reporting on the court hearing](#) confirmed that the India options strategy was revealed inadvertently by Millennium's own counsel — who apologized upon doing so — and that a Jane Street attorney stated the strategy "was one of the firm's most lucrative." Judge Paul Engelmayer confirmed on record that Jane Street claimed the strategy generated approximately \$1 billion in 2023 profits.

[Filings reviewed by BNN Bloomberg](#) confirmed additional details: Jane Street claimed its profits from the strategy fell by more than 50% in March 2024, which it attributed exclusively to Millennium beginning to deploy the same approach. Jane Street also stated it had spent "tens of millions of dollars" over multiple years developing the strategy, and that the firm relies on confidentiality agreements rather than non-compete clauses to protect its intellectual property. A key passage from the amended complaint states the strategy was "counterintuitive" and "built upon the years-long effort and findings of the algorithmic team."

The strategy's geography — Indian equity index options, primarily BANKNIFTY and Nifty 50 contracts on the National Stock Exchange of India — made it visible to Indian regulators. As [documented in public](#)

[filings](#), the case revealed the extraordinary scale and concentration of Jane Street's derivatives trading in India, attracting the attention of SEBI. In July 2025, SEBI formally accused Jane Street of manipulating the BANKNIFTY and Nifty 50 index settlement prices and ordered ₹4,843 crore — approximately \$566 million — in alleged profits placed into escrow, as [confirmed by Reuters](#).

The litigation's deepest relevance for market making analysis is structural: it publicly established that a single options strategy — concentrated in a single national derivatives market — generated \$1 billion in profits in one year for a firm that makes markets across thousands of instruments globally. The case settled before its mechanics were fully disclosed. The docket remains [Jane Street Group LLC v. Millennium Management LLC, 24 cv 02783, SDNY](#).

The Millennium settlement did not close the legal record on Jane Street's market making operations. In February 2026, the court-appointed administrator winding down Terraform Labs filed suit against Jane Street in Manhattan federal court ([Case No. 1:26-cv-1504](#)), alleging that in May 2022, a wallet linked to Jane Street withdrew \$85 million in TerraUSD from a decentralized liquidity pool within ten minutes of Terraform quietly withdrawing \$150 million from the same pool — before any public announcement — and that this action contributed to TerraUSD losing its dollar peg and triggering the \$40 billion Terra/Luna collapse. On April 23, 2026, Jane Street filed a motion to dismiss, calling the lawsuit "a transparent attempt to extract cash" and arguing that its largest

trades occurred after damaging information about TerraUSD was already public. Jane Street denies all allegations. The case is active. Its relevance here is not the outcome — which is not yet determined — but the pattern: for the third time in two years, a federal court is being asked to determine exactly where Jane Street's information advantage in financial markets ends and unlawful conduct begins. That question is the same question Glosten-Milgrom formalized in 1985. It has not been answered definitively in law.

Optiver's CFTC Record: Where Market Making Mechanics Become Settlement Manipulation

The Jane Street case drew that regulatory line in options markets. The clearest documentation of where market makers have historically crossed it in futures markets — specifically by exploiting the mechanics of daily price settlement — is the Optiver CFTC enforcement record.

The [CFTC press release dated April 20, 2012](#), covering a consent order entered by Chief Judge Loretta A. Preska of the SDNY, documented that Optiver and its traders engaged in 19 separate instances of manipulation over 11 days in March 2007, targeting NYMEX crude oil, heating oil, and gasoline futures. The mechanism: Optiver deployed a rapid-fire execution tool internally nicknamed "The Hammer" to execute large volumes of orders during the final minutes of trading, moving the settlement price — the price at which futures contracts are marked for daily P&L and final

delivery — in the direction of pre-existing positions. The settlement included a \$13 million civil monetary penalty plus \$1 million in disgorgement, totaling \$14 million.

The structural lesson is not primarily about the manipulation itself but about what it reveals regarding the mechanics of settlement-adjacent trading. Futures settlement prices are calculated from actual trades in a defined closing window. A market maker or principal trader with substantial derivatives positions has an inherent financial interest in where settlement falls. When the aggregate position is large enough and the settlement window is thin enough, the economics can favor trading in the underlying to move settlement — even at a loss on those trades — because the gain on the derivatives book outweighs the cost.

The CFTC's Director of Enforcement, David Meister, stated:

"Manipulative schemes like 'banging the close' harm market integrity."

The regulatory line between aggressive market making near settlement — legal — and manipulation of the settlement calculation itself — illegal — is precisely the line that the Jane Street-SEBI dispute re-opened in India in 2025, and that the Terraform Labs lawsuit re-opens in a federal court in New York in 2026, at scales that Optiver's 2007 manipulation case did not approach.

The Three Structural Levers: A Unified Framework

The enforcement cases — Citadel's SIP arbitrage, Optiver's banging the close, Jane Street's India options concentration — collectively define the outer boundary of permissible market making. What the voluntary disclosures and academic record show is the mechanism operating well inside that boundary.

Every documented market making P&L — Virtu's 1,237-out-of-1,238-day record, XTX's £1.1 billion profit year, Jane Street's India billion, Citadel's 35% retail flow dominance — decomposes into combinations of three structural levers.

Lever 1: Fair-Value Model Accuracy

The primary alpha source in prediction-quality market making is the accuracy of the firm's estimate of where the asset will trade over its intended hold horizon. As [XTX's own public client documentation](#) states, the firm's pricing edge is "cross-asset pricing alpha — the ability to successfully forecast short-term price movements." This is not vague marketing language. It is a direct statement that the bid and ask prices XTX quotes are functions of a multi-asset model that predicts near-term price direction, and that the profitability of the strategy depends on whether those predictions are correct.

Lever 2: Adverse Selection Identification

The second lever is the identification and segregation of informed flow. Every disclosed market making operation operates some version of a toxicity filter. The [Virtu CORRESP filing](#) describes their real-time risk system: "if our risk management system detects a trading strategy generating revenues outside of our preset limits, it will freeze, or lockdown that strategy and alert management." At a trade-population level, this is automated detection of strategy-level adverse selection — when a quoting strategy begins generating unexpected losses, the system interprets this as evidence of informed counterparties and shuts the strategy down. Clients whose flow is systematically toxic are offered different, wider pricing; those with balanced, uninformed flow receive tighter spreads.

Lever 3: Inventory Velocity and Hedging

The third lever is how quickly accumulated inventory is neutralized. The Avellaneda-Stoikov framework handles this via quote skewing. In practice, inventory neutralization takes two forms: passive — waiting for the other side of the trade to arrive at skewed prices — and active — hedging in a related instrument. The [Virtu CORRESP filing](#) describes the active form precisely: "Virtu hedges with related securities when they accumulate inventory so that they would end the day flat in terms of risk but not necessarily in terms of their position." The distinction between risk neutrality — no net Greeks or delta exposure — and position

neutrality — flat shares — is the operative difference between market making and directional trading.

 **If this analysis interests you, the full deep-dive is on Patreon:**

[Virtu Made Money 1,237 Out of 1,238 Trading Days. Here Is Exactly How — Documented in SEC Filings, Federal Court Records, and CFTC Orders Filed by Virtu, XTX, Jane Street, Citadel, and Optiver Themselves](#)

This piece documents how each of the three levers is implemented in practice — with exact quotes from Virtu's SEC CORRESP filing, Zar Amrolia's on-record Risk.net interviews, and the Avellaneda-Stoikov reservation price derivation sourced to the original Quantitative Finance paper.

The Capacity Ceiling: Primary Evidence That Edges Erode

Market making alpha compresses structurally as competition intensifies, and the public record provides specific evidence of this compression.

Virtu's own SEC filings trace the compression trajectory. The [2017 8-K earnings filing](#) shows that average daily Adjusted Net Trading Income fell to approximately \$1.175 million per day in Q2 2017, down from \$1.599

million in Q2 2016 — a 26.5% decline in revenue per day despite trading in more than 12,000 financial instruments across 235 venues in 36 countries. Revenue declined across all categories: Americas equities fell 18.5%, global commodities fell 28.8%, global currencies fell 29.0%. This is spread compression at global scale, documented quarterly in SEC filings.

XTX's response to capacity pressure was geographic and product expansion combined with the zero-hold-time strategic pivot. After moving to zero hold time, XTX grew its direct client base from 20 to 100 in 12 months — trading per-trade margin compression for volume scale. Jeremy Smart, XTX's head of distribution, is quoted directly in the December 2017 Risk.net awards profile: "The jump in volume and clients has got a lot to do with people realising that having a zero hold time shows a clear demonstration of our commitment to transparency and fairness, which gives us extra weighting." The strategic logic: when spread per trade falls, the competitive response is to win more trades — a volume-scale trade-off that only works if operating costs do not scale linearly with volume, which they do not in an automated market making operation.

The regulatory dimension of capacity compression is the most structurally uncertain. Payment for order flow was banned by the EU under [MiFIR amendments that entered into force on March 28, 2024](#), subject to a transitional phase-out period extending to June 30, 2026 for certain member states. India's SEBI actions against Jane Street in 2025

introduced regulatory uncertainty around structured expiry-day options positioning in developing derivatives markets. The [CFTC's FY2023 enforcement results](#) documented 96 enforcement actions and \$4.3 billion in penalties — evidence that the regulatory cost of running near the line of permissible market making conduct has become materially non-zero.

What the Record Shows and What It Doesn't

The primary sources reviewed here collectively establish six claims with documentary specificity:

1. **Virtu** ran 1,238 trading days with one loss — meaning 1,237 profitable days — winning just 50.4% of individual trades, through diversification across 11,000+ instruments in 35 countries. *Source: Virtu S-1 (2014) and CORRESP filing, SEC.*
2. **XTX** holds positions 10–20 minutes on average in FX, explicitly choosing prediction quality over latency, and documented 50% bilateral volume growth and a client base expansion from 20 to 100 by removing last-look protection. *Source: Risk.net, on-record interviews, 2017.*

3. **Citadel's** FastFill and SmartProvide used a SIP-vs-direct-feed latency gap to internalize retail orders at prices better for Citadel than its stated commitment required, while processing approximately 35% of average daily US retail equity volume. Citadel settled for \$22.6 million. *Source: SEC Administrative Proceeding Release No. 33-10280.*
4. **Jane Street** ran a single India options strategy that generated approximately \$1 billion in 2023 profits, experienced a greater than 50% profit decline when a competitor replicated the strategy, and spent tens of millions of dollars developing it. *Source: SDNY Case 1:24-cv-02783-PAE, confirmed in open court.*
5. **VPIN**, the primary real-time adverse selection metric, was developed by a practitioner at Tudor Investment Corp and explicitly proposed as a live market maker risk management tool. *Source: Journal of Portfolio Management, Vol. 37, No. 2, 2011.*
6. **Optiver's** "banging the close" scheme involved 19 manipulation instances across 11 days in March 2007, resulting in a \$14 million total settlement — \$13 million civil monetary penalty plus \$1 million disgorgement. *Source: CFTC consent order, April 2012.*

What the record does not show is the specific algorithmic content of any of these strategies. The Jane Street complaint's key sections are redacted. The Citadel order describes behavior but not model architecture. XTX's volatility surface parameterization, cross-asset signal weights, and toxicity classification thresholds remain proprietary. The public record

establishes what these firms do and what it produces. How they do it at the model level is the trade secret the litigation was trying to protect.

For a quant PM evaluating the category: the documented evidence is sufficient to conclude that market making alpha is real, demonstrable over multi-year periods in public filings, and decomposable into three identifiable technical dimensions. The hard part is that the infrastructure requirements — co-location across 35+ countries, sub-millisecond risk monitoring, real-time cross-asset pricing models for 50,000+ instruments — represent capital and engineering barriers that compound with time. The firms whose records are documented here have been building those barriers since 2009. The public filings show the results. The source code is not in any of them.

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This piece contains every primary source linked inline at the point of the claim — SEC filings, CFTC consent orders, federal court complaints, and peer-reviewed academic papers — so every number is traceable to its named document without leaving the page.

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